

Supplemental Material: Exact minimum measurement dependence for violating the CHSH inequality

[Author name]¹

¹[Affiliation]

This Supplement contains (A) the table of even-parity product patterns, (B) details of the exact-curve certificate, (C) the complete verification record with re-run instructions, and (D) the literature sweep underlying the novelty statements. All certificates are self-contained Python scripts using exact rational or $\mathbb{Q}(\sqrt{2})$ arithmetic (no floating point); each exits with code 0 if and only if every assertion passes.

I. PRODUCT PATTERNS

The eight even-parity patterns $q = (q_{00}, q_{01}, q_{10}, q_{11})$ with $q_{00}q_{01}q_{10}q_{11} = +1$, their CHSH values $C(q) = q_{00} + q_{01} + q_{10} - q_{11}$, and their relation to the sign vector $\sigma = (+, +, +, -)$:

Pattern	$C(q)$	#diff. from σ	Name	Misses
$(-1, +1, +1, -1)$	+2	1	M_A	$q_{00} = +1$
$(+1, -1, +1, -1)$	+2	1	M_B	$q_{01} = +1$
$(+1, +1, -1, -1)$	+2	1	M_C	$q_{10} = +1$
$(+1, +1, +1, +1)$	+2	1	M_D	$q_{11} = -1$
$(+1, -1, -1, +1)$	-2	3	—	(negative of M_A)
$(-1, +1, -1, +1)$	-2	3	—	(negative of M_B)
$(-1, -1, +1, +1)$	-2	3	—	(negative of M_C)
$(-1, -1, -1, -1)$	-2	3	—	(negative of M_D)

TABLE I. The eight even-parity product patterns. Each miss-type pattern differs from σ at exactly one coordinate; the parity lemma follows because σ itself has product -1 .

II. EXACT-CURVE CERTIFICATE

Theorem 5 of the main text (exact curves for $N = 2, \dots, 8$) is certified by exhaustive evaluation of the finite-maximum formula

$$S_{\max}(N, F) = \max_Q [S_0(Q) + \sum_{ab} g_{ab}(F)], \quad (1)$$

over all $\binom{8+N-1}{N}$ multisets Q of even-parity patterns, in exact rational arithmetic:

For $N \geq 9$ only the first segment $S_{\max} = 2 + 8F$ on $[0, 1/N]$, the cap $\lceil N/4 \rceil / N$, and the headline value $S_{\max}(N, F^*) = 2\sqrt{2}$ are proved; the full bend region is left to numerics (the enumeration cost grows as $\binom{N+7}{7}$; $N = 9$ already requires 11 440 multisets, feasible but not performed in this work).

N	$\binom{8+N-1}{N}$	bend point(s)	$S_{\max}$ at bends	cap	$F(S=4; N)$
2	36	—	—	—	$\frac{1}{4}$
3	120	—	—	—	$\frac{1}{3}$
4	330	—	—	—	$\frac{1}{2}$
5	792	$F = \frac{1}{5}$	$\frac{18}{5}$	$\frac{2}{5}$	$\frac{2}{5}$
6	1716	$F = \frac{1}{6}$	$\frac{10}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
7	3432	$F = \frac{1}{7}$	$\frac{22}{7}$	$\frac{2}{7}$	$\frac{2}{7}$
8	6435	—	—	—	$\frac{1}{4}$

TABLE II. Enumeration sizes and certified bend values. The upper envelope of the exact affine functions agrees with the round-robin lower bound on every cell; slopes are in $\{0, 2, 4, 6, 8\}$ as the formula forces.

III. VERIFICATION RECORD AND CODE AVAILABILITY

Every result in the main text is checkable without trusting the prose. Three independent routes are provided.

A. One-sided: exact row formula and case analysis

The one-sided row optimum of Lemma 3 of the main text has the exact piecewise-linear form

$$\begin{aligned}
 h(f, F) &= \mu + G^*(F), \quad G^*(F) = \max_{\delta \in [0, F]} [P(\delta) + N(\delta)], \\
 P(\delta) &= 2 \min(\delta, k(N-1)/N) \\
 &\quad - 2 \max(0, \delta - k(N-1)/N - z(N-1)/N), \\
 N(\delta) &= 2 \min(\delta, m/N) \\
 &\quad - 2 \max(0, \delta - m/N - z/N),
 \end{aligned} \quad (2)$$

where $f \in \{-2, 0, +2\}^N$ has k entries $+2$, z entries 0 , m entries -2 , and $\mu = 2(k-m)/N$. The form is water-filling: with $d = p - \rho$ and $\delta = \sum_{d>0} d_\lambda \leq F$, the positive part of d (per-entry cap $1 - 1/N$) is placed at rate $+2$ on the k entries and then at rate 0 on the z entries, giving $P(\delta)$; the negative part (per-entry cap $1/N$) is placed at rate $+2$ on the m entries (mass removed from a -2 entry gains $+2$) and then at rate 0 , giving $N(\delta)$; overlapping placements cancel L^1 norm, never gain. The script cross-checks Eq. (2) against an independent LP for all (k, z, m) types, $N = 2, \dots, 8$, on a dense F -grid including every breakpoint (4109 pairs, maximum deviation 1.1×10^{-15}).

The bound $S_{\text{os}}(N, F) \leq 2 + 4F$ of Proposition 2 of the main text follows by applying the lemma to the two rows. Write $Z_A := \mu_A + 2F + 2 \min(F, m_A/N)$ and Z_B likewise,

where μ_A, μ_B and m_A, m_B refer to f^A, f^B ; then $X := h(f^A, F) = \min(2, Z_A)$, $Y = h(f^B, F) = \min(2, Z_B)$, $S \leq X + Y$, and by the key structural fact $k_A + m_A + k_B + m_B = N$ with $\mu_A + \mu_B = 2 - 4(m_A + m_B)/N$. *Case (a)* $Z_A, Z_B \leq 2$: $S \leq Z_A + Z_B = 2 - 4(m_A + m_B)/N + 4F + 2\min(F, m_A/N) + 2\min(F, m_B/N) \leq 2 + 4F$. *Case (b)* $Z_A > 2 \geq Z_B$: $X = 2$ and one needs $Z_B \leq 4F$, i.e. $(k_B - m_B)/N + \min(F, m_B/N) \leq F$. From $Z_A > 2$: $(k_A - m_A)/N + F + \min(F, m_A/N) > 1$; since $-m_A/N + \min(F, m_A/N) \leq 0$, this gives $k_A > N(1 - F)$, hence $k_B = N - k_A - m_A - m_B < NF - m_A - m_B$ and $(k_B - m_B)/N + \min(F, m_B/N) < F - (m_A + 2m_B)/N + \min(F, m_B/N) \leq F$, so $S \leq 2 + Z_B \leq 2 + 4F$. *Case (c)* is symmetric to (b). *Case (d)* $Z_A, Z_B > 2$: adding the two strict inequalities gives $(k_A + k_B - m_A - m_B)/N + 2F + \min(F, m_A/N) + \min(F, m_B/N) > 2$, but the left side is $\leq 1 - (m_A + m_B)/N + 2F < 2$ for $F < \frac{1}{2}$ (using $k_A + k_B \leq N - (m_A + m_B)$ and $\min(F, m_A/N) \leq m_A/N$); for $F \geq \frac{1}{2}$ one has $X = Y = 2$ and the trivial bound $S \leq 4 \leq 2 + 4F$ suffices.

The status of the exact one-sided curves certified by `onesided.py` is: for even N , the single-segment form $S_{\text{os}, \max}(N, F) = \min(2 + 4F, 4)$ is proved for *all* even N (upper: the case analysis plus $S \leq 4$; lower: the Q_N^* construction); for odd $N = 2M + 1$, the two-segment form $2 + 4F$ on $[0, M/N]$, $(6M + 2)/(2M + 1) + 2F$ on $[M/N, (M + 1)/N]$, then 4, is certified by exact enumeration for $N = 3, 5, 7$ and is conjectured for odd $N \geq 9$; the cap $F_{\text{os}}(S=4; N) = \lceil N/2 \rceil / N$ is proved for all N : $S = 4$ forces each row onto its $f^A = +2$ (resp. $f^B = +2$) support, costing $\text{TV} \geq (N - k_A)/N$ and $(N - k_B)/N$ with $k_A + k_B \leq N$, and the round-robin split attains the maximum.

B. Exact certificates (no floating point)

- `audit1.py` $\rightarrow$ `audit1.out` — rational arithmetic. **Part A:** the exact $S_{\text{max}}(N, F)$ curves and cap minimum over *all* multisets for $N = 2, \dots, 8$ (Theorems 3 and 5 of the main text; Theorem 4 follows from Theorem 2 for $N \geq 4$ and from Theorem 5 for $N = 2, 3$, and is float-cross-checked in Part B). **Part B:** floating-point cross-check of F^* at $N = 2, \dots, 6$: $S_{\text{max}}(N, F_N^*) = 2.828427124746$ to 12 digits (margins to the runner-up construction: 0 for $N = 2, 3, 5, 6$ — multiple optimal Q 's; 2.07×10^{-1} for $N = 4$, where Q_4 is the unique optimiser). **Part C:** residuals of the saved witness arrays, attributed to the bisection tolerance (10^{-5}) of the search pipeline. **Part D:** the $N = 4$ witness table of Sec. 6 of the main text as an *exact identity* in $\mathbb{Q}(\sqrt{2})$: $E[ab] = (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2})$, $S = 2\sqrt{2}$, per-row $\text{TV} = F^* = (\sqrt{2} - 1)/4$ exactly. **Part E:** the two-state NS certificate (Theorem 6 of the main text): all 256 response vertices, union-find closure over the four NS equations, 1312 candidate affine functions, every one ≤ 2 on $F \in [0, \frac{1}{2}]$.

- `ns_general.py` $\rightarrow$ `ns_general.out` — Theorem 7 (NS headline) certificates. **Part A:** the $N \geq 4$ closed form verified exactly — row constraints and all four NS identities with d formal, for $N = 4, \dots, 64$; plus the feasibility inequality $\lfloor N/4 \rfloor \geq (\sqrt{2} - 1)N/4$ for $N = 4, \dots, 999$. **Part B:** the $N = 3$ witness in $\mathbb{Q}(\sqrt{2})$: $S = 2\sqrt{2}$ exactly, per-row TV $(d_3, d_3, d_3/2, d_3)$, all four NS residuals 0. **Part C:** the closed forms versus the saved witness arrays (maximum deviation 5.6×10^{-17}).
- `prbox_ns.py` $\rightarrow$ `prbox_ns.out` — Theorem 8 (NS algebraic maximum) certificates; 28 checks, exit code 0. **Part A:** exact rational recomputation of the signalling cap $\min_Q \max_{ab} n_{ab}(Q) = \lceil N/4 \rceil$ over *all* even-parity multisets for $N = 2, \dots, 8$ (self-contained lower bound, with the monotonicity $S_{\text{NS}} \leq S_{\text{max}}$). **Part B:** the closed-form construction of Sec. V of the main text rebuilt from the formula in exact rational arithmetic for $N = 3, \dots, 300$ plus spot checks at $N = 1000, 4096, 10000$: row sums 1, non-negativity, $E[ab] = \sigma_{ab}$ (hence $S = 4$), per-row $\text{TV} \leq \lceil N/4 \rceil / N$, all four NS residuals 0; explicit witnesses printed for $N = 3, 4, 5$ (the $N = 4$ witness is an explicit local decomposition of the PR box at cost $1/4$). **Part C:** independent HiGHS LP (float) feasibility at $F = \lceil N/4 \rceil / N$ for $N = 3, \dots, 16$ — a search cross-check only; the claims rest on Parts A–B.
- `onesided.py` $\rightarrow$ `onesided.out` — Proposition 2 (one-sided measurement dependence) certificates. **Part 0:** the exact row formula Eq. (2) cross-checked against an independent LP for all (k, z, m) types, $N = 2, \dots, 8$ (4109 pairs, maximum deviation 1.1×10^{-15}). **Part A:** the exact $S_{\text{os}, \max}(N, F)$ curves for $N = 2, \dots, 8$ (exact upper hull over all $\binom{8+N-1}{N}$ multisets), asserting the closed form of Sec. III A, including $S_{\text{os}, \max}(N, 0) = 2$ and first-segment slope 4. **Part B:** the headline $F_{\text{os}}^*(N) = (\sqrt{2} - 1)/2$ as an exact $\mathbb{Q}(\sqrt{2})$ identity, including $(\sqrt{2} - 1)/2 < \lfloor N/2 \rfloor / N$ for $N = 2, \dots, 8$, and the cap $\lceil N/2 \rceil / N$. **Parts C1–C3:** the one-sided-NS variants (not cited in the main text): $S_{\text{NS}, \text{os}}(2, F) \equiv 2$ and $S_{\text{NS}, \text{os}}(3, F) \equiv 2$ at all F , and $F_{\text{NS}, \text{os}}^*(N) = (\sqrt{2} - 1)/2$ for $N \geq 4$ with explicit $\mathbb{Q}(\sqrt{2})$ witnesses for $N = 4, \dots, 8$.
- `metric_compare.py` $\rightarrow$ `metric_compare.out` — the metric comparison of Sec. VII.5 of the main text. Exact $\mathbb{Q}(\sqrt{2})$ parts: per-row TVs, all six pairwise TVs, M_{model} , M_{full} , and NS residuals of the four closed-form witnesses; the saturation check $M_{\text{model}} = 2F^*$ for the enumerated witnesses and the $N = 3$ -NS non-saturation ($M_{\text{model}} = (\sqrt{2} - 1)/2 = \frac{3}{2}F^*$); the converse $F \leq M_{\text{model}} + \text{TV}(p_{\text{avg}} \parallel \rho)$ and the fixed-baseline counterexample

(all rows $q \neq \rho$ gives $M_{\text{model}} = 0$ with $F > 0$); Hall’s upper bound $S \leq \min\{2 + 3M, 4\}$ re-proved without the no-signalling assumption. **Part G** (high-precision numerics, 60 decimal digits, stated as such): $I(X, Y; \Lambda) = 0.056036$ bits for both $N = 4$ witnesses, 0.054810 for the $N = 3$ NS witness, against the Hall–Branciard floors $I_R \approx 0.046274$ and $I_C \approx 0.080170$ recomputed from their formulas.

- **fringe_pm.py** $\rightarrow$ **fringe_pm.out**, Part PM-7 — Proposition 3 (aggregation robustness) of the main text: the $N = 4$ witness arithmetic redone in exact $\mathbb{Q}(\sqrt{2})$ — all four per-row TVs equal to $d = (\sqrt{2} - 1)/4$ exactly, $S = 2\sqrt{2}$ exactly, hence the average-TV cost is d ; independent HiGHS LP cross-check minimising the average TV at the known-optimal vertex subject to $S \geq 2\sqrt{2}$ (returns d to $< 10^{-7}$). The full script contains 88 checks; the remainder certifies the Peres–Mermin results of the companion paper (in preparation).

- **chk2_superdet_garza.py** $\rightarrow$ **out_superdet_garza.txt** — the Garza–Hance figures of Sec. VII.4 of the main text: single-band TV $\sin \theta / (\pi \Delta L)$ (analytic vs numerical integration, $< 1.3 \times 10^{-7}$); the mixture TV $s(\pi - 2s) / (\pi(\pi - s))$ at $\Delta L = 0.77$ (error 1.66×10^{-8} vs numerical integration); self-consistency of their correlator ($\max_{\theta} |E(\theta; \Delta L)| \leq 1$ holds exactly at $\Delta L = \frac{1}{2}$ and fails just below it, 1.0315 at $\Delta L = 0.45$); $|S| \approx 2.82610$ at best fit; the bound $S \leq 2 + 8F$ over full parameter grids (margin $\geq -10^{-9}$). *Flag*: these are floating-point evaluations of the *authors’ published formulas* (the θ values are transcendental, so exact rational/ $\mathbb{Q}(\sqrt{2})$ arithmetic does not apply) — a cross-check of an external model, not a certificate of the theorems of the main text.

C. Independent LP cross-checks (floating point)

- **verify.py** $\rightarrow$ **verify_run.log** (and byte-identical rerun **verify_rerun.log**): 21 checks pass, exit code 0 — the $N = 4$ witness by direct arithmetic; the Bell bound $S_{\text{max}}(F=0) = 2$ for $N = 2, 3, 4$; the curve formula at 12 (N, F) points; F^* bisection for $N = 2, 3, 4$.
- **capregion.out**: the LP reproduces the exact bend values — $S_{\text{max}}(5, \{0.2, 0.25, 0.3\}) = 3.6/3.7/3.8 (= \frac{16}{5} + 2F)$ to $\leq 1.8 \times 10^{-15}$; $S_{\text{max}}(6, 0.2) = \frac{8}{3} + 4(0.2)$

exactly; the caps $S_{\text{max}}(5, 0.4) = S_{\text{max}}(6, \frac{1}{3}) = 4$ by explicit construction.

- **ns_tight_N{3,4}.out**: direct evaluations $S_{\text{NS}}(3, (\sqrt{2} - 1)/3) = S_{\text{NS}}(4, (\sqrt{2} - 1)/4) = 2\sqrt{2}$ to 4.4×10^{-16} — the exact Theorem 7 values. Earlier bisection offsets in older logs (≈ 0.13812 , ≈ 0.10358) were resolution artifacts of a 13-iteration bisection on $[0, 0.5]$ (resolution 6.1×10^{-5}), not evidence of an NS gap.
- **ns_witness.out**: the saved NS witnesses re-verified in a fresh process from raw arrays — S to 4.4×10^{-16} , per-row TV $\leq F^* + 10^{-9}$ with at least one row exactly at F^* , all four NS residuals $\leq 5.6 \times 10^{-17}$.

The witness arrays are checked independently of the pipeline that searched for them (recomputation from raw (p, u, v) arrays with plain arithmetic), and the exact certificates are a third route, independent of both.

D. Code availability

The scripts **audit1.py**, **ns_general.py**, **verify.py**, **prbox_ns.py**, **onesided.py**, **metric_compare.py**, **fringe_pm.py**, **chk2_superdet_garza.py**, and the LP pipeline (**phase1.py**–**phase3.py**), together with all output logs and witness arrays, are available from the authors on request. [TODO: replace with a public repository URL before submission.]

IV. LITERATURE SWEEP

The novelty statements of Sec. VII of the main text are a search result with stated coverage limits, not a claim of proven novelty. The sweep covered arXiv-indexed literature on measurement dependence and Bell nonlocality from 2015 through 2026-08-23: ~ 16 papers were read at full or abstract depth (the rest at catalogue level); pay-walled journal content not mirrored on arXiv, non-arXiv repositories, and books/theses are not covered. The complete record is in **literature_check.out**, available from the authors on request alongside the other logs: its Section A verifies the anchor works (Hall 2010; Hall & Branciard 2020; Bacciagaluppi, Hermens & Leegwater 2026) and a separate Task-3 section verifies Friedman et al. 2019, all against primary sources; Section B lists the candidate related works found in the sweep with a per-paper overlap verdict (including the four closest: Alai 2026 $\times 2$, Pal et al. 2026, Takakura et al. (arXiv:2208.13634; Quantum 9, 1662 (2025))); Section C states the novelty verdict and these coverage limits; Section D is the search log (queries actually run).